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Effects of water and rice straw management practices on water savings and greenhouse gas emissions from a double-rice paddy field in the Central Plain of Thailand

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ABSTRACT

Rice-rice (double-rice) cropping in the Central Plain region of Thailand relies heavily on irrigation water inputs for cultivation under flooded conditions, which is increasingly constrained by water scarcity. In this production system, farmers often burn or incorporate rice straw (RS) in the field. The effects of the combined use of water and RS management on irrigated lowland double-rice cropping, with respect to irrigation water savings, grain yields, and greenhouse gas (GHG) emissions, remain largely unknown. Field experiments of two consecutive rice-growing seasons in 2016–2017 were conducted at the Ayutthaya Rice Research Center in Ayutthaya, Thailand, to investigate the effects of different water and RS management practices on irrigation water savings, rice grain yields, and GHG emissions. The treatments consisted of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD; 15 cm threshold water level below the soil surface for irrigation]) and three RS management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]). Parameters such as irrigation water inputs, soil characteristics (pH, temperature, water content), above-ground biomass, and grain yields were measured. The static vented flux chamber technique was used to measure GHG emissions (CH_4 and N_2O fluxes). AWD irrigation significantly reduced water inputs compared to CF in both growing seasons; however, there were more water savings in the dry season. AWD also resulted in higher grain yields compared to CF in both growing seasons. The average CH_4 emissions from WRS plots were significantly lower compared to both RS-I and RS-B plots, regardless of the growing seasons. However, RS-I and RS-B plots under AWD had 39% and 63% lower CH_4 emissions, respectively, in the wet season, while the corresponding values were 37% and 36%, respectively, in the dry season compared to the average CH_4 emissions from these RS treatments under CF. Water and RS management did not significantly influence the average flux of N_2O emissions from paddy fields in the wet season; however, CF had a lower average N_2O flux than AWD in the dry season. The average flux of N_2O emissions were very low compared to the CH_4 emissions flux in both seasons. Averaged across two seasons, the yield-scaled global warming potential (GWP) under AWD was reduced by 34% compared to CF, and under WRS it was reduced by 36% and 29% compared to RS-I and RS-B, respectively. RS application, either as soil incorporation or open-field burning, had little or no effect on irrigation water savings and grain yields; however, the yield-scaled GWP was significantly higher from plots with RS-I or RS-B than WRS plots. Among the water regime and RS management practices tested, AWD irrigation without any RS application (soil incorporation or burning) can be employed for maintaining the yield-scaled GWP of irrigated lowland double-rice cropping at a low level.

1. Introduction

Agriculture is one of the most important sectors of Thailand's economy. Thailand is one of the world's major producers and exporters of rice (*Oryza sativa* L.) (Ullah et al., 2018a). The nation's food security is also dependent on the agricultural sector, especially rice. Thailand's

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Central Plain region contributes the largest share (30%) to the national rice production (Maneepitak et al., 2019). In Thailand, rice is cultivated in two growing seasons: the rainy/wet season (May–October) and the summer/dry season (November–April). Farmers in the Central Plain region cultivate a major portion of the dry season rice (~47% of the total production) in irrigated fields (OAE, 2018). Drought and poor soil fertility are the major constraints to upland rice cultivation in Thailand, while water scarcity and pest infestation largely limit irrigated lowland rice production (Ullah et al., 2018b).

Agricultural production is a substantial source of global greenhouse gas (GHG) emissions (Burney et al., 2010). Levels of emissions are subject to cultivation practices and processes throughout the agricultural production phases and the product utilization. Irrigated lowland rice production systems are known to be a significant source of methane (CH_4) and nitrous oxide (N_2O). Field water management, fertilization, organic matter application (crop residue, manure), and sowing methods affect CH_4 and N_2O emissions from paddy fields (Zou et al., 2005; Wang et al., 2012). CH_4 formation from the flooded rice field occurs through a methanogenesis process strictly under anaerobic conditions, during which organic matter undergoes decomposition and CH_4 is the metabolic end-product (Demirel and Scherer, 2008). Agricultural soil management activities largely contribute to the production of N_2O through regulating the rate and direction of nitrification and denitrification processes (Malla et al., 2005).

Rice grown under flooded conditions requires substantial inputs of freshwater and almost 80% of Asia's existing water resources are devoted to irrigated lowland rice production (Bouman et al., 2007; Ullah et al., 2017). Freshwater resources available to farmlands are rapidly declining due to climate change and heavy extraction of groundwater posing a serious threat to the sustainability of traditional flooded rice cultivation (Jabran et al., 2015). More rice needs to be produced with less water to boost food supply for the growing world population. Therefore, alternative rice production methods must be sought to reduce irrigation water inputs and increase water productivity (Belder et al., 2004; Datta et al., 2017; Ullah and Datta, 2018). Among various water-efficient techniques, alternate wetting and drying (AWD) is the most widely promoted water-saving irrigation method to cope with the increasing threat of water scarcity in irrigated lowland rice production (Belder et al., 2004; Datta et al., 2017; Ullah and Datta, 2018; Ullah et al., 2019). AWD irrigation has the potential to reduce irrigation water inputs and CH_4 emissions into the atmosphere while also maintaining or increasing grain yields (Belder et al., 2004; Watanabe et al., 2013; Chu et al., 2015; Linquist et al., 2015). For example, a reduction of 15–20% in water use has been reported in AWD-irrigated lowland rice cultivation without any yield reduction (Belder et al., 2004). Also, LaHue et al. (2016) reported a reduction of 57–78% in annual CH_4 emissions from rice fields maintained under the AWD system. However, N_2O emissions might increase under AWD irrigation due to an enhancement in nitrification and denitrification processes resulting from constantly changing soil conditions between anaerobic and aerobic and related changes in redox potential (Towprayoon et al., 2005; Oo et al., 2018). Despite the many visible benefits, large-scale adoption of AWD in irrigated flooded rice production systems is not evident in Thailand, including the Central Plain region, due to farmers' fears of possible yield reductions.

Intensive double-rice cropping in the Central Plain region of Thailand generates about 10 million tonnes of straw residues each year (OAE, 2018; Maneepitak et al., 2019). Thai rice farmers manage their straw predominantly through burning *in situ* and soil incorporation. Soil incorporation of rice straw (RS) positively influences soil fertility and the ecological environment (Wang et al., 2015a, b). Open-field burning is popular in Thailand, where about 69% of the produced straw is burnt *in situ*. This is because the method is cost-effective, helping to reduce pest populations, quickly clearing the field before initiating mechanized land preparation, and making the land available for the next rice crop (OAE, 2018; Maneepitak et al., 2019). However, soil incorporation of RS has been promoted by the Land Development Department and the Pollution Control Department to reduce atmospheric pollution caused by open-field burning. Burning of stubble has a direct negative impact on soil fertility, soil organic matter, and the environment by causing air pollution and GHG emissions (Chu et al., 2015; Wang et al., 2015a, b; Wang et al., 2016). Annual GHG emissions from open-field burning of rice residue in Thailand ranged from 12,274 to 43,693 Gg CO_2-eq, or almost 2% of GHG emissions from the nation's agricultural activity (Cheewaphongphan et al., 2011). It has been reported that RS incorporation increases CH_4 emissions from paddy fields (Zhang et al., 2011; Yuan et al., 2014; Wang et al., 2015b). Straw incorporation-induced N_2O emissions draw mixed reviews, with both an increase (Zhang et al., 2013) and a decrease (Zou et al., 2005) in N_2O emissions having been reported. Usually, an increase in CH_4 emissions due to the incorporation of RS may counteract the occasional decrease in N_2O emissions from paddy fields (Zhang et al., 2011; Yuan et al., 2014). RS incorporation could increase CH_4 emissions from paddy fields due to the presence of anaerobic conditions caused by waterlogging (Wang et al., 2016). Therefore, further evaluation of the emissions dynamics of these two important GHGs under different water and RS management practices is needed. Straw incorporation-driven GHG emissions from paddy fields largely depend on water management practices and climatic conditions (Wang et al., 2015b). Although burning of RS is less labor-intensive than soil incorporation, field burning results in substantial air pollution and associated health risks (Kutcher and Malhi, 2010).

The double-rice cropping system in Thailand, especially in the Central Plain region, has been challenged by inefficient use of freshwater resources and poor management of RS. Irrigation water savings, grain yields, and GHG emissions of irrigated lowland double-rice cropping under different water regimes and RS management practices have not been thoroughly investigated. It is, therefore, important to identify suitable combinations of water and residue management for double-rice cropping in the agro-ecological environment of the Central Plain region. We hypothesized that proper combinations of water and residue management practices would reduce water inputs, increase or maintain yields, alter the GHG emissions dynamics, and decrease overall global warming potential (GWP) of the double-rice cropping system. The objective of the present study was to evaluate the effects of water regimes and RS management practices on irrigation water savings, grain yields, and GHG emissions (CH_4 and N_2O) of the irrigated lowland double-rice cropping system practiced in the Central Plain region of Thailand.

2. Materials and methods

2.1. Experiment site

The field experimental site was located at the Ayutthaya Rice Research Center (14°21′54.79″N, 100°36′19.71″E, 2 m above mean sea level) in Ayutthaya, Thailand. This region has a tropical savanna climate characterized by warm temperatures throughout the year, and two distinct wet (May–October) and dry (November–April) seasons. The mean annual temperature is around 27 °C and the annual rainfall ranges from 1000–1400 mm. Almost all of the year's rainfall occurs during the wet season. The soil of the experimental field is derived from marine sediment-mixed riverine alluvium, having a slow runoff rate and a ground water depth of > 2 m. The paddy soil is strongly acidic (very-fine, mixed, active, acid isohyperthermic, Vertic Endoaquepts) and is classified as Ayutthaya soil series. The soil is a poorly-drained clay with the main soil properties (0–15 cm depth) as follows: sand 14%, silt 22%, clay 64%, pH (1:1 soil-water) 6.0, organic matter 1.41%, total carbon (C) (w/w) 0.95%, total nitrogen (N) (w/w) 0.16%, available phosphorus (P) 17 mg kg^{-1}, available potassium (K) 201 mg kg^{-1}, available calcium (Ca) 5129 mg kg^{-1}, available magnesium (Mg) 953 mg kg^{-1}, and electrical conductivity (EC) 0.81 dS m^{-1}. Double-rice

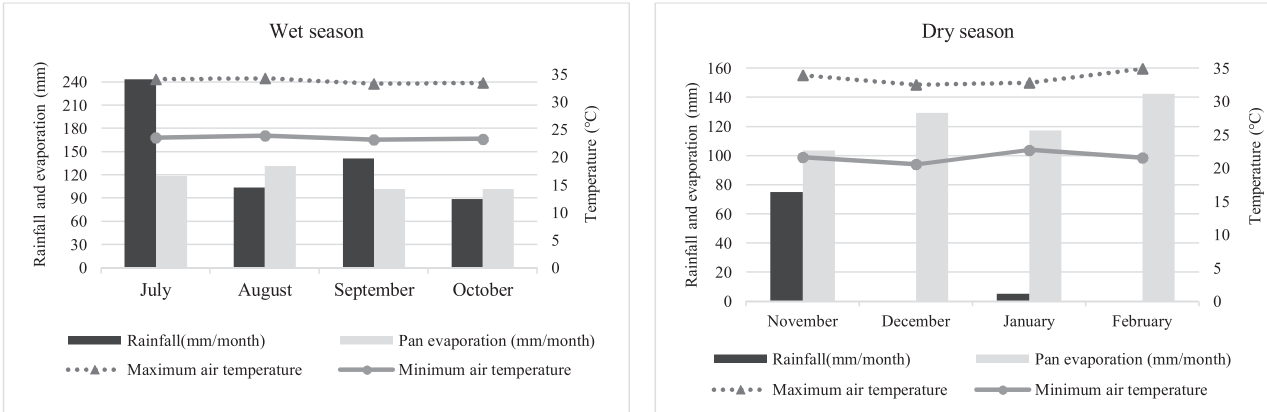

Fig. 1. Mean monthly maximum and minimum air temperatures, total monthly rainfall, and evaporation at the Ayutthaya Rice Research Center in Ayutthaya, Thailand, during the wet and dry rice growing seasons of 2016–2017.

cropping has been in practice at the experimental site for more than 10 years. Maximum and minimum air temperatures as well as total monthly rainfall and evaporation data during the rice growing seasons of 2016–2017 are presented in Fig. 1.

2.2. Experimental design and agronomic management

Two field experiments were conducted during the wet (July–October 2016) and dry (November 2016–February 2017) seasons of 2016–2017. The treatments followed a split-plot design with water management as the main plot and RS management as the subplot. The main plot included two water management practices: continuous flooding (CF) and alternate wetting and drying (AWD), whereas the subplot consisted of three RS management practices: RS incorporation (RS-I), RS burning (RS-B), and without RS incorporation and burning (WRS). There were three replications of each treatment. The individual plot size was 35 m^2 (7 m × 5 m) and plots were separated by a 1 m-wide alley. Bunds (30 cm height) were constructed along each side of the plot and were covered with black plastic film inserted to a depth of 30 cm below the soil surface to prevent the lateral water movement. Besides separating plots by the 1 m-wide alley, additional bunds and plastic film were used around the plots to minimize the lateral movement of water among the subplots. Water was pumped from a nearby field ditch into the center of each main plot. Irrigation water input in each plot was measured by a flow meter installed in the irrigation pipeline. The mode and timing of experimental field management activities in the rice fields (RS management, crop establishment, fertilizer application, and water management) under CF and AWD irrigation during the wet and dry growing seasons are presented in Table 1.

The rice variety Pathumthani 1 was used for both growing seasons. This is a photoperiod-insensitive variety, and the maturity period ranges between 106 and 126 days depending on the cultivation method (Ullah and Datta, 2018; Ullah et al., 2018a,b). Seeds of Pathumthani 1 were sown by the pre-germinated broadcasting method (wet direct seeding) into the main field in both seasons (Table 1). Pre-germinated seed broadcasting is a common practice in the study area. However, farmers usually keep the field under a continuously flooded condition during most of the growing season, and the burning of rice straw *in situ* is a popular method of straw disposal. The same synthetic fertilizer was applied in all plots in both growing seasons. Insect pests, diseases, and weeds were chemically controlled to avoid yield loss by following the recommended plant protection measures for rice cultivation in Ayutthaya Province.

2.3. Collection and measurement of CH_4 and N_2O fluxes

The static vented flux chamber technique was used to measure CH_4 and N_2O fluxes (Chu et al., 2015). Each gas sampling chamber consisted of an acrylic cover resting on a stainless steel base with an opening at the bottom. The chamber includes a permanent base that was inserted into the soil in such a way that rice plants can grow inside. Extension of the permanent base to varying lengths is possible to accommodate the growing plants. The chamber also contains a lid, which is equipped with a vent tube, a fan, and a thermocouple wire. The base, made of a polyvinyl chloride (PVC) frame with dimensions of 60 × 60 × 60 cm (width × length × height), was positioned at least 1.5 m from the edge of the plot and was inserted into the soil to a depth of 15 cm. During the period of gas sampling, the permanent base was covered with the lid.

In the wet season, the fluxes of CH_4 and N_2O were measured from each plot every seven days, starting from 8 to 26 days after broadcasting (DAB) (seedling stage) and including 64–105 DAB (flowering stage), by collecting air samples in triplicate, whereas gas samples were collected every three days between 26 and 64 DAB (tillering stage). In the dry season, the fluxes of CH_4 and N_2O were measured from each plot every seven days, starting from 13 to 27 DAB (seedling stage) and including 61–112 DAB (flowering stage), whereas gas samples were collected every three days between 27 and 61 DAB (tillering stage). Sampling was conducted in the morning between 0900 and 1100 h, as the soil temperature closely approximates the mean daily soil temperature during this period (Zou et al., 2005; Xu et al., 2015). A fan installed on the top of the chamber was turned on for about 30 s before gas sampling. This was to ensure proper mixing (homogenization) of gases inside the chamber. Soil and air temperatures inside the chamber were recorded for each set of emissions measurements. Gas samples were collected at equal time intervals of 0, 5, 10, 15, and 20 min after covering using a 25 ml polypropylene syringe. These samples were analyzed within a few hours of sampling using a gas chromatograph (GC, Agilent 6890, Agilent Technologies, Wilmington, DE, USA) equipped with a flame ionization detector (FID) for detecting CH_4 at 350 °C and an electron capture detector (ECD) for detecting N_2O at 375 °C. The average fluxes and standard errors (SEs) of CH_4 and N_2O were calculated from three replications. Total (seasonal) amounts of CH_4 and N_2O emissions under each water regime and RS management practice were calculated from the daily measurement. The flux of gas emissions was calculated using the following equation (Peng et al., 2011):

$$F = \frac{10^{-5} \times \mu P}{R(T+273.2)} \times H \times \frac{dc}{dt}$$

where F is the CH_4 or N_2O emissions flux (mg m^{-2} h^{-1}), μ is the gas molecular weight (CH_4 = 16.123 g mol^{-1}, N_2O = 44.013 g mol^{-1}), P is the mean air pressure in the chamber (P = 1.01325 × 10^5 Pa for low elevation area), R is the universal gas constant (R = 8.31441 J mol^{-1} K^{-1}), T is the mean air temperature in the chamber (°C), H is the

Table 1
Mode and timing of experimental field management activities in rice field during the wet and dry growing seasons of 2016–2017 under two water regimes (continuous flooding [CF] and alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]).

Management activities	Wet season		Dry season	
	CF	AWD	CF	AWD
Land preparation				
RS-I	RS from the preceding season was soaked in water for two days and was manually spread onto the soil surface at the rate of 5000 kg ha^{-1}. This treatment was applied ˜ two weeks before broadcasting the rice seeds in both seasons followed by tillage operations with a power tiller			
RS-B	RS was distributed and burned under open-field conditions at the rate of 5000 kg ha^{-1}. A butane-fueled lighter was used to ignite RS and the burning typically lasted for 3–5 min in each plot. This treatment was applied ˜ two weeks before broadcasting the rice seeds in both seasons			
WRS	There was no RS incorporation or burning in WRS plots			
Crop establishment	Pathumthani 1 variety was sown by pre-germinated broadcasting method (wet direct seeding) into the main field on 5 July and harvested on 18 October 2016. The seed rate was 125 kg ha^{-1}		Seeds of the same rice variety were broadcasted (wet direct seeding) on 31 October 2016 and harvesting was done on 26 February 2017. The seed rate was 125 kg ha^{-1}	
Fertilizer application	Fertilizer was applied at the rate of 156 kg ha^{-1} at 21 and 24 days after broadcasting (DAB) in the wet and dry seasons, respectively, in the form of NPK (16:20:0) as basal. Urea (46:0:0) was top-dressed at the rate of 94 kg ha^{-1} at 65 and 69 DAB in the wet and dry seasons, respectively			
Irrigation	The plot was continuously flooded by maintaining a constant surface water depth of 1–2 cm during 1–14 DAB, 5 cm water during 15–92 DAB, and irrigation was stopped two weeks before harvest at 106 DAB	The plot was flooded with 1–2 cm water from the first day of broadcasting until 25 DAB, and then irrigation was stopped. The plot was re-irrigated to a depth of ˜ 5 cm once the water level above the soil surface dropped to 15 cm below the soil surface as indicated by a perforated field-water tube installed in each plot. This AWD cycle continues until 60 DAB (initial flowering stage)	The plot was flooded with 1–2 cm water during 1–14 DAB, 5 cm water during 15–105 DAB, and irrigation was stopped two weeks before harvest at 119 DAB	The plot was flooded with 1–2 cm water from the first day of broadcasting until 30 DAB, and then irrigation was stopped. The plot was re-irrigated to a depth of ˜ 5 cm once the water level above the soil surface dropped to 15 cm below the soil surface as indicated by a perforated field-water tube installed in each plot. This AWD cycle continues until 61 DAB (initial flowering stage)

effective height of the static chamber corresponding to the difference between the height of the static chamber and the depth of the water layer in the field when sampling (cm), and d*c*/d*t* is the increase rate of gas concentration in the chamber (the units of CH_4 and N_2O are ppm h^{-1} and ppb h^{-1}, respectively).

The seasonal total CH_4 or N_2O emissions from each plot were calculated by linearly interpolating the gas emissions during the sampling dates, assuming the measured fluxes represent the average daily fluxes. Cumulative emissions were calculated according to the following equation, as described by Li et al. (2013):

$$CE = \sum \frac{F_i + F_{i+1}}{2} \times 10^{-3} \times d \times 24$$

where *CE* is the seasonal total emissions (kg ha^{-1}), F_i and F_{i+1} are the measured fluxes of two consecutive sampling days (mg m^{-2} h^{-1}), and d is the number of days between two consecutive sampling days (d).

2.4. Global warming potential (GWP) and yield-scaled GWP

The GWP based on the CH_4 and N_2O emissions was used to account for the climatic impact of paddy rice cultivation in the wet and dry seasons under two water regimes and three RS management practices. The total CH_4 and N_2O emissions from each of the treatments were converted to CO_2 equivalent (CO_2-eq.) emissions. The combined GWP over a 100-year time span from each of the treatments was calculated using the following equations (IPCC, 2013):

$$GWP_{CH_4} = \text{Seasonal total } CH_4 \text{emissions} \times 34$$

$$GWP_{N_2O} = \text{Seasonal total} N_2O \text{ emissions} \times 298$$

$$GWP_{total} = GWP_{CH_4} + GWP_{N_2O}$$

where GWP_{total} is the combined CO_2-eq. GWP (kg CO_2-eq. ha^{-1}) of CH_4 (GWP_{CH_4}) and N_2O (GWP_{N_2O}). The yield-scaled GWP (kg CO_2-eq. kg^{-1} grain yield) from each of the treatments was calculated by dividing the growing season's total CO_2-eq. GWP (GWP_{total}) by per kg of rice grain produced.

Yield-scaled GWP (kg CO_2-eq. kg^{-1} grain yield) = GWP_{total} / Rice grain yield (kg)

2.5. Additional measurements

The mean monthly maximum and minimum temperatures as well as total monthly rainfall and evaporation data were collected from the Ayutthaya Rice Research Center. In both seasons, the water volume input from irrigation to each plot was measured using a flow meter installed in the irrigation pipeline. Soil pH was measured in the laboratory immediately after the sampling using a pH meter. Soil temperature at 5 cm depth was measured from each treatment while measuring the gas emissions using a soil thermometer. The temperature inside the chamber was also recorded during each gas sampling using the thermometer installed inside the chamber. Soil samples were collected, weighted, oven-dried, and reweighted, and then the gravimetric water content was measured by dividing the mass of water (i.e., the loss in weight) to the oven dry weight of soil, and expressed as g g^{-1}. Volumetric water content was determined by multiplying the gravimetric water content with the bulk density of the soil, and expressed as cm^3 cm^{-3}.

Rice plants were hand-harvested at maturity from a 5 m × 2 m area in each plot (the plants under the chamber were not included), and the grain yield was measured. Plants were harvested from the middle section of each plot to reduce any border effects. At the same time, 10 hills of plant samples from each plot were randomly sampled and oven-dried at 70 °C until constant weight to determine above-ground biomass, and expressed as kg ha^{-1}. The final grain yields from all treatments were adjusted to 14% moisture content.

Table 2
Irrigation water inputs and irrigation water savings during the wet and dry rice growing seasons of 2016–2017 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]).

Factor	Wet season		Dry season	
	Irrigation water inputs ($m^3\ ha^{-1}$)	Irrigation water savings compared to CF (%)	Irrigation water inputs ($m^3\ ha^{-1}$)	Irrigation water savings compared to CF (%)
Water management (W)				
CF	9636 ± 591.29a		11903 ± 487.16a	
AWD	6632 ± 196.03b	31	6895 ± 274.26b	42
RS management (R)				
RS-I	8250 ± 354.68		9416 ± 162.48	
RS-B	8029 ± 230.51		9313 ± 200.44	
WRS	8123 ± 505.80		9467 ± 779.22	
W × R				
CF + RS-I	10056 ± 633.36		12090 ± 162.38	
CF + RS-B	9522 ± 544.13		11969 ± 307.90	
CF + WRS	9329 ± 596.38		11650 ± 991.20	
AWD + RS-I	6444 ± 75.99	36	6742 ± 162.58	44
AWD + RS-B	6536 ± 96.88	31	6658 ± 92.97	44
AWD + WRS	6916 ± 415.22	26	7284 ± 567.23	37
F-value				
W	65.45**		154.56**	
R	0.12ns		0.05ns	
W × R	0.87ns		0.64ns	

Data are means ± standard errors of three replications. In a column, numbers followed by same letters are not significantly different by Tukey's honest significant difference test at $P < 0.05$. ns: not significant.
** indicates significance at $P < 0.01$.

2.6. Statistical analysis

Data were analyzed using the statistical package SPSS 20.0 (SPSS Inc., Chicago, IL, USA). A two-way ANOVA for split-plot design was conducted to evaluate the effects of water regimes, RS management practices, and their interactions on irrigation water inputs, soil parameters, grain yields, above-ground biomass, average and seasonal fluxes of CH_4 and N_2O emissions, total GWP, and yield-scaled GWP. Means for significant treatment effects were separated by Tukey's honest significant difference test. Differences in the mean values at $P < 0.05$ were considered statistically significant.

3. Results

3.1. Irrigation water inputs and water savings

The two-way interaction between water regime and RS management was not significant ($P > 0.05$) for irrigation water inputs in both growing seasons; however, the individual effect of the water regime was highly significant (Table 2). The plots maintained under CF had significantly higher irrigation water inputs, which was reduced by 31% and 42% under AWD in the wet and dry seasons, respectively. AWD saved 36%, 31%, and 26% of irrigation water inputs for RS-I, RS-B, and WRS plots, respectively, compared to CF for the same RS management practices in the wet season. Irrigation water savings was 44%, 44%, and 37% for RS-I, RS-B, and WRS plots, respectively, compared to CF in the dry season.

3.2. Soil pH, soil temperature, soil water content, and rice production

The individual effect of water regime and RS management was highly significant ($P < 0.01$) for soil pH in both growing seasons (Table 3). In the wet season, the mean soil pH was higher under CF (7.28) than under AWD (7.22). RS-B plots had a lower mean soil pH (7.17) than RS-I (7.28) and WRS (7.29) plots in the wet season. A similar trend was also observed in the dry season in which AWD had a significantly lower soil pH (7.08) than CF (7.13), and RS-B plots had a significantly lower soil pH (7.03) than both RS-I (7.13) and WRS (7.16) plots. The mean soil temperature and soil water content were similar among treatments in both seasons, and their changes over time did not significantly vary among treatments.

Grain yield was highly significantly ($P < 0.01$) affected by water regime in both growing seasons (Table 4). Overall, grain yield was higher in plots maintained under AWD than under CF in both seasons. Grain yield was 15% and 7% higher under AWD compared to CF in the wet and dry seasons, respectively. Above-ground biomass was similar among treatments in the wet season, whereas it was highly significantly affected by RS management in the dry season (Table 4). In the dry season, RS-I plots had 41% and 42% lower above-ground biomass than RS-B and WRS plots, respectively.

3.3. Seasonal dynamics of CH_4 and N_2O emissions

Water regime and RS management had an influence on the average flux of CH_4 and N_2O emissions from the paddy field in both growing seasons (Table 4). Seasonal variations in the CH_4 and N_2O fluxes measured from each plot in the wet and dry seasons are presented in Fig. 2. The two-way interaction between water regime and RS management was highly significant ($P < 0.01$) for the average flux of CH_4 in the wet season, whereas the individual effect of RS management was highly significant ($P < 0.01$) in the dry season. The average emissions rate of CH_4 in the wet season was higher than the dry season. In the wet season, the highest average flux of CH_4 emissions ($7.34\ mg\ m^{-2}\ h^{-1}$) was obtained from RS-B plots under CF, whereas it was the lowest ($2.73\ mg\ m^{-2}\ h^{-1}$) for the same RS management under AWD. In the dry season, CH_4 emissions were significantly lower from WRS plots ($1.30\ mg\ m^{-2}\ h^{-1}$) compared to RS-I ($4.32\ mg\ m^{-2}\ h^{-1}$) and RS-B ($2.89\ mg\ m^{-2}\ h^{-1}$) plots. The average flux of CH_4 emissions was similar between RS-I and RS-B plots in the dry season. Similar seasonal patterns of CH_4 emissions were observed from all three RS management plots under each water regime (Fig. 2). In the wet season, the CH_4 emissions flux peaked at $21.62\ mg\ m^{-2}\ h^{-1}$ at 36 DAB from RS-B plots under CF. RS-B plots resulted in the smallest peaks ($0.63\ mg\ m^{-2}\ h^{-1}$) at 43 DAB under AWD. In the dry season, the CH_4 emissions flux peaked at $21.89\ mg\ m^{-2}\ h^{-1}$ at 20 DAB from RS-I plots under CF, whereas the same treatment had the smallest peaks ($0.11\ mg\ m^{-2}\ h^{-1}$) at 61 DAB under AWD.

The two-way interaction between water regime and RS management as well as the individual effect was not significant ($P > 0.05$) for the average flux of N_2O emissions in the wet season; however, the individual effect of water regime was highly significant in the dry season (Table 4). The average emissions rate of N_2O was slightly higher in the wet season than in the dry season. In the wet season, RS-B plots exhibited peaks as high as $0.19\ mg\ m^{-2}\ h^{-1}$ under AWD, whereas RS-I plots resulted in peaks as low as $0.16\ mg\ m^{-2}\ h^{-1}$ under CF. In the dry season, N_2O emissions were significantly higher under AWD ($0.15\ mg\ m^{-2}\ h^{-1}$) compared to CF ($0.13\ mg\ m^{-2}\ h^{-1}$). Overall, there was a reduction in N_2O emissions under CF compared to AWD. The N_2O emissions flux peaked at $0.50\ mg\ m^{-2}\ h^{-1}$ at 77 DAB from RS-B plots under CF, whereas RS-I plots had the smallest peaks ($0.01\ mg\ m^{-2}\ h^{-1}$) at 43 DAB under CF in the wet season (Fig. 2). In the dry season, the N_2O emissions flux peaked at $0.25\ mg\ m^{-2}\ h^{-1}$ at 55 DAB from WRS plots under AWD, whereas WRS and RS-I plots exhibited the smallest peaks ($0.06\ mg\ m^{-2}\ h^{-1}$) at 13 and 43 DAB, respectively, under CF.

3.4. Seasonal GWP and yield-scaled GWP

The two-way interaction between water regime and RS management had a highly significant ($P < 0.01$) effect on the total GWP in the

Table 3
Mean soil pH, soil temperature, and soil water content during the wet and dry rice growing seasons of 2016–2017 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]).

Factor	Wet season			Dry season		
	Soil pH	Soil temperature (°C)	Soil water content ($cm^3\ cm^{-3}$)	Soil pH	Soil temperature (°C)	Soil water content ($cm^3\ cm^{-3}$)
Water management (W)						
CF	7.28 ± 0.03a	28.39 ± 0.92	0.36 ± 0.03	7.13 ± 0.02a	23.63 ± 1.26	0.49 ± 0.02
AWD	7.22 ± 0.04b	28.35 ± 0.90	0.33 ± 0.03	7.08 ± 0.02b	23.68 ± 1.34	0.49 ± 0.01
RS management (R)						
RS-I	7.28 ± 0.04a	28.50 ± 0.91	0.34 ± 0.04	7.13 ± 0.02a	24.19 ± 1.60	0.49 ± 0.02
RS-B	7.17 ± 0.04b	28.04 ± 0.91	0.34 ± 0.04	7.03 ± 0.02b	24.02 ± 1.54	0.49 ± 0.02
WRS	7.29 ± 0.04a	28.56 ± 0.75	0.34 ± 0.03	7.16 ± 0.03a	22.76 ± 0.62	0.48 ± 0.02
W × R						
CF + RS-I	7.32 ± 0.04	28.69 ± 0.96	0.34 ± 0.04	7.16 ± 0.02	22.87 ± 0.57	0.48 ± 0.02
CF + RS-B	7.24 ± 0.03	27.79 ± 0.83	0.35 ± 0.05	7.05 ± 0.02	25.31 ± 2.58	0.50 ± 0.02
CF + WRS	7.29 ± 0.03	28.68 ± 0.96	0.38 ± 0.01	7.17 ± 0.02	22.72 ± 0.63	0.48 ± 0.01
AWD + RS-I	7.25 ± 0.04	28.31 ± 0.85	0.34 ± 0.03	7.11 ± 0.02	25.50 ± 2.62	0.50 ± 0.01
AWD + RS-B	7.11 ± 0.04	28.30 ± 0.99	0.33 ± 0.02	7.00 ± 0.02	22.72 ± 0.49	0.48 ± 0.01
AWD + WRS	7.30 ± 0.04	28.44 ± 0.85	0.30 ± 0.04	7.14 ± 0.03	22.81 ± 0.60	0.49 ± 0.02
F-value						
W	3.75**	0.01ns	0.79ns	4.13**	0.01ns	0.08ns
R	5.27**	0.19ns	0.01ns	12.15**	0.48ns	0.11ns
W × R	1.57ns	0.14ns	0.57ns	0.02ns	1.36ns	0.40ns

Data are means ± standard errors of three replications. In a column, numbers followed by same letters are not significantly different by Tukey's honest significant difference test at $P < 0.05$. ns: not significant.
** indicates significance at $P < 0.01$.

Table 4
Rice grain yields, above-ground biomass, and average flux of CH_4 and N_2O emissions from the paddy field during the wet and dry rice growing seasons of 2016–2017 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]).

Factor	Wet season				Dry season			
	Grain yield ($kg\ ha^{-1}$)	Above-ground biomass ($kg\ ha^{-1}$)	CH_4 ($mg\ m^{-2}\ h^{-1}$)	N_2O ($mg\ m^{-2}\ h^{-1}$)	Grain yield ($kg\ ha^{-1}$)	Above-ground biomass ($kg\ ha^{-1}$)	CH_4 ($mg\ m^{-2}\ h^{-1}$)	N_2O ($mg\ m^{-2}\ h^{-1}$)
Water management (W)								
CF	3710 ± 182.45b	17892 ± 3560.93	5.41 ± 0.79a	0.17 ± 0.02	3910 ± 60.08b	11601 ± 1419.33	3.39 ± 1.00	0.13 ± 0.01b
AWD	4281 ± 157.60a	17560 ± 3344.49	2.99 ± 0.39b	0.17 ± 0.02	4195 ± 60.56a	13253 ± 1780.75	2.81 ± 0.67	0.15 ± 0.01a
RS management (R)								
RS-I	3889 ± 162.89	19064 ± 4644.03	4.54 ± 0.59a	0.16 ± 0.02	4063 ± 63.82	8399 ± 1063.38b	4.32 ± 1.35a	0.14 ± 0.01
RS-B	4122 ± 98.17	19322 ± 3251.87	5.04 ± 0.82a	0.18 ± 0.02	4096 ± 23.36	14335 ± 1767.23a	2.89 ± 0.77a	0.14 ± 0.01
WRS	3975 ± 249.02	14791 ± 2462.23	3.02 ± 0.37b	0.18 ± 0.02	3999 ± 93.79	14547 ± 1313.01a	1.30 ± 0.21b	0.15 ± 0.01
W × R								
CF + RS-I	3591 ± 221.31	19116 ± 4925.18	5.64 ± 0.69ab	0.16 ± 0.02	3856 ± 60.95	6897 ± 751.67	5.31 ± 1.64	0.13 ± 0.01
CF + RS-B	4048 ± 65.14	19074 ± 3078.77	7.34 ± 1.33a	0.17 ± 0.02	4049 ± 26.19	15034 ± 1703.70	3.52 ± 0.83	0.14 ± 0.01
CF + WRS	3491 ± 260.91	15486 ± 2678.83	3.25 ± 0.36bc	0.18 ± 0.02	3825 ± 93.10	12871 ± 1802.62	1.35 ± 0.17	0.14 ± 0.01
AWD + RS-I	4187 ± 104.46	19013 ± 4362.87	3.44 ± 0.49bc	0.17 ± 0.02	4270 ± 66.69	9901 ± 1375.09	3.33 ± 1.06	0.14 ± 0.01
AWD + RS-B	4196 ± 131.20	19569 ± 3424.97	2.73 ± 0.31c	0.19 ± 0.02	4142 ± 20.53	13636 ± 1830.76	2.26 ± 0.71	0.15 ± 0.01
AWD + WRS	4460 ± 237.13	14096 ± 2245.63	2.79 ± 0.38c	0.17 ± 0.02	4173 ± 94.47	16224 ± 2136.41	1.25 ± 0.24	0.16 ± 0.01
F-value								
W	14.27**	0.01ns	18.84**	0.04ns	27.27**	1.49ns	2.20ns	3.94**
R	0.81ns	1.01ns	4.72**	0.38ns	1.08ns	8.85**	5.40**	1.27ns
W × R	2.46ns	0.04ns	4.68**	0.32ns	3.20ns	1.27ns	0.53ns	0.38ns

Data are means ± standard errors of three replications. In a column, numbers followed by same letters are not significantly different by Tukey's honest significant difference test at $P < 0.05$. ns: not significant.
** indicates significance at $P < 0.01$.

wet season (Table 5). In the wet season, RS-B plots had the largest total GWP (6533.94 kg CO_2-eq. ha^{-1}), followed by RS-I plots (5505.24 kg CO_2-eq. ha^{-1}) under CF, while WRS plots had the lowest total GWP (3380.86 kg CO_2-eq. ha^{-1}) under AWD. In the dry season, the individual effect of water and RS management was highly significant for the total GWP (Table 6). The total GWP was significantly lower under AWD (2886.94 kg CO_2-eq. ha^{-1}) compared to CF (3648 kg CO_2-eq. ha^{-1}). Similarly, RS-I plots had the largest total GWP (4264.86 kg CO_2-eq. ha^{-1}), followed by RS-B plots (3350.00 kg CO_2-eq. ha^{-1}), whereas WRS plots had the lowest total GWP (2187.60 kg CO_2-eq. ha^{-1}). The total GWP in the wet season was decreased by 45%, 27%, and 12% from RS-B, RS-I, and WRS plots under AWD irrigation, respectively, compared to the same RS management practice under CF (Table 5). In the dry season, RS-I plots had a 27% and 95% higher total GWP compared to RS-B and WRS plots, respectively (Table 6).

The two-way interaction between water regime and RS management was not significant ($P > 0.05$) for the yield-scaled GWP in both growing seasons; however, the individual effect of water regime and RS management was highly significant (Tables 5 and 6). In the wet season, the yield-scaled GWP was 64% higher under CF than AWD (Table 5). Similarly, WRS plots had the lowest yield-scaled GWP (0.93 kg CO_2-eq. kg^{-1} grain yield), which was 26% and 25% lower than RS-I and RS-B

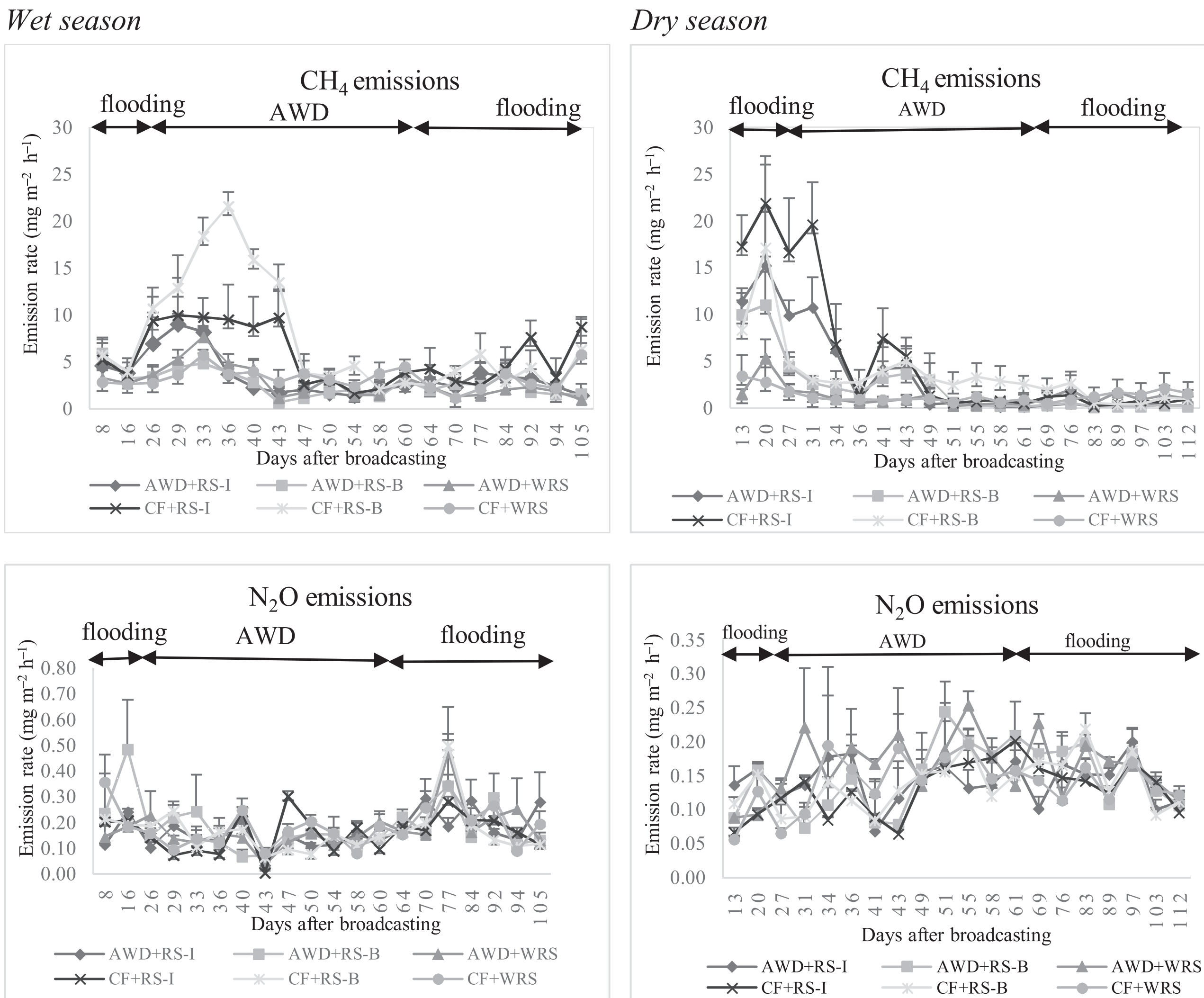

Fig. 2. Variations of CH_4 and N_2O fluxes over time from the paddy field during the wet and dry rice growing seasons of 2016–2017 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]).

plots, respectively. A similar trend was also observed in the dry season in which CF had a 35% higher yield-scaled GWP than AWD, and RS-I plots had a 29% and 93% higher yield-scaled GWP than RS-B and WRS plots, respectively (Table 6).

4. Discussion

4.1. Effects of water regime and RS management on water inputs and rice production

In this study, irrigation water inputs was significantly decreased under AWD compared to CF, regardless of the growing seasons. AWD saved 31% and 42% of irrigation water inputs compared to CF in the wet and dry seasons, respectively. AWD has been developed to reduce irrigation water inputs without resulting in adverse effects on rice production (Linquist et al., 2015; Datta et al., 2017; Ullah and Datta, 2018). Alternative means of rice production such as the direct seeding method of cultivation (dry or wet) and AWD irrigation are becoming increasingly popular among farmers in many Asian countries in the context of increasing irrigation water scarcity (Ullah et al., 2017; Ullah and Datta, 2018). A 53% reduction in irrigation water use has been reported in direct-seeded rice when water is applied through intermittent irrigation at four day intervals without any yield reduction (Talebnejad and Sepaskhah, 2014). Lagomarsino et al. (2016) reported between 40% and 70% water saving under AWD compared to CF in dry-seeded rice. In Thailand, a water savings of 20–50% has been recorded under AWD compared to CF, and this variation in water savings has been attributed to soil type and weather variability (ONEP, 2015). The results of the present study are in line with these findings. Reduced water inputs and high water productivity under AWD irrigation have been attributed to reduced seepage and percolation, along with reduced evapotranspiration (Cabangon et al., 2004).

Soil pH was slightly reduced under AWD compared to CF in both growing seasons. Mean soil temperature and soil water content did not significantly differ among the treatments in both seasons. Lagomarsino et al. (2016) also observed no marked differences in soil and water characteristics between CF and AWD plots. Overall, there was no marked difference in grain yields between the two growing seasons, and the plants maintained under AWD had a higher grain yield compared to CF, regardless of the seasons. Setyanto et al. (2018) observed a significant difference in grain yields of rice in the wet and dry seasons subjected to AWD and CF irrigation, and attributed that difference to various planting methods and weather conditions between the two seasons. This contradicts the present findings in which we observed similar grain yields between the two seasons. This similarity between the two growing seasons might be due to the same seeding method,

Table 5
Seasonal total CH_4 and N_2O emissions from the paddy field during the wet rice growing season of 2016 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]), and their contributions to the global warming potential (GWP) and yield-scaled GWP.

Factor	Total CH_4 emissions (kg ha^{-1})	Total N_2O emissions (kg ha^{-1})	GWP of CH_4 (kg CO_2-eq. ha^{-1})	GWP of N_2O (kg CO_2-eq. ha^{-1})	Total GWP (kg CO_2-eq. ha^{-1})	Yield-scaled GWP (kg CO_2-eq. kg^{-1} grain yield)
Water management (W)						
CF	118.25a	4.25	4020.50a	1265.51	5286.00a	1.41a
AWD	68.82b	4.46	2339.77b	1330.07	3669.84b	0.86b
RS management (R)						
RS-I	103.95a	4.11	3534.30a	1223.29	4757.59a	1.25a
RS-B	109.78a	4.51	3732.35a	1343.98	5076.33a	1.24a
WRS	66.88b	4.45	2273.75b	1326.10	3599.85b	0.93b
W × R						
CF + RS-I	126.86b	4.00	4313.24b	1192.00	5505.24ab	1.53
CF + RS-B	155.10a	4.23	5273.40a	1260.54	6533.94a	1.61
CF + WRS	72.79c	4.51	2474.86c	1343.98	3818.84bc	1.09
AWD + RS-I	81.04c	4.21	2755.36c	1254.58	4009.94bc	0.96
AWD + RS-B	64.45c	4.79	2191.30c	1427.42	3618.72c	0.86
AWD + WRS	60.96c	4.39	2072.64c	1308.22	3380.86c	0.76
F-value						
W	146.62**	0.25ns	146.62**	0.25ns	38.37**	70.01**
R	43.32**	0.35ns	43.32**	0.35ns	11.82**	10.10**
W × R	31.26**	0.21ns	31.26**	0.21ns	7.57**	3.31ns

In a column, numbers followed by same letters are not significantly different by Tukey's honest significant difference test at $P < 0.05$. ns: not significant.
** indicates significance at $P < 0.01$.

Table 6
Seasonal total CH_4 and N_2O emissions from the paddy field during the dry rice growing season of 2016–2017 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]), and their contributions to the global warming potential (GWP) and yield-scaled GWP.

Factor	Total CH_4 emissions (kg ha^{-1})	Total N_2O emissions (kg ha^{-1})	GWP of CH_4 (kg CO_2-eq. ha^{-1})	GWP of N_2O (kg CO_2-eq. ha^{-1})	Total GWP (kg CO_2-eq. ha^{-1})	Yield-scaled GWP (kg CO_2-eq. kg^{-1} grain yield)
Water management (W)						
CF	78.93a	3.24	2683.51a	964.53	3648.03a	0.93a
AWD	53.62b	3.57	1823.08b	1063.86	2886.94b	0.69b
RS management (R)						
RS-I	96.65a	3.29	3285.93a	978.93	4264.86a	1.06a
RS-B	68.34b	3.45	2323.39b	1026.61	3350.00b	0.82b
WRS	33.84b	3.48	1150.56b	1037.04	2187.60b	0.55b
W × R						
CF + RS-I	118.49a	3.18	4028.66a	947.64	4976.30	1.29
CF + RS-B	85.11b	3.32	2893.74b	989.36	3883.10	0.96
CF + WRS	33.18d	3.21	1128.12d	956.58	2084.70	0.55
AWD + RS-I	74.80bc	3.39	2543.20bc	1010.22	3553.42	0.83
AWD + RS-B	51.56cd	3.57	1753.04cd	1063.86	2816.90	0.68
AWD + WRS	34.50d	3.75	1170.00d	1117.50	2287.50	0.54
F-value						
W	38.43**	0.46ns	38.43**	0.46ns	7.14**	12.05**
R	79.14**	0.06ns	79.14**	0.06ns	17.82**	17.79**
W × R	11.15**	0.05ns	11.15**	0.05ns	3.01ns	3.65ns

In a column, numbers followed by same letters are not significantly different by Tukey's honest significant difference test at $P < 0.05$. ns: not significant.
** indicates significance at $P < 0.01$.

similar weather pattern, and soil conditions. An increase in grain yield and a reduction in water use under AWD irrigation have been attributed to: (i) reduction of unproductive tillers, (ii) removal of toxic substances, (iii) prevention of root rot, and (iv) enhancement in root system development (Islam et al., 2018; Ullah and Datta, 2018). Soil drying during AWD is an important factor for positively influencing the grain yield of rice as it provides optimum growth conditions for root systems, remobilizes C reserves from vegetative tissues to grains, and enhances the grain filling rate (Chu et al., 2015; Setyanto et al., 2018).

The reported effect of AWD on grain yield is not consistent, but highly variable. Some authors have observed a decrease in rice grain yield under AWD (Li et al., 2011; Linquist et al., 2015; Xu et al., 2015; Lagomarsino et al., 2016), whereas others have reported no yield reduction (Belder et al., 2004; Dong et al., 2012; Yao et al., 2012; Pandey et al., 2014; Carrijo et al., 2018; Setyanto et al., 2018), or even an increase in yield (Liu et al., 2013; Ullah et al., 2018b). This variable yield response to AWD among studies is attributed to variation in the range of water management regimes (LaHue et al., 2016; Ullah and Datta, 2018), degree of AWD severity (Zhang et al., 2009; Linquist et al., 2015), timing (Linquist et al., 2015; LaHue et al., 2016), rice variety (Xu et al., 2015), soil moisture monitoring methodology (Zhang et al., 2009; Lampayan et al., 2015; Linquist et al., 2015), and weather and soil hydrological conditions of the experimental site (Tuong et al., 2005; Dong et al., 2012). Zhang et al. (2009) reported an increase in grain yield (11%) from a field experiment in China when sandy loam soils were allowed to dry to soil water potential at −15 kPa prior to rewetting, but there was a decrease in grain yield (32%) when fields were allowed to dry to soil water potential at −30 kPa before rewetting. Bouman et al. (2007) suggested practicing "safe" AWD to avoid a yield reduction in which a water level of 15 cm (soil water potential at

–10 to −15 kPa) in perforated field water tubes installed into the soil could be used as the threshold for rewatering in most growth stages of rice. This index for rice irrigation has been supported by many field observations, where an increase in water use efficiency and grain yields have been reported (Dong et al., 2012; Yao et al., 2012; Pandey et al., 2014; Chu et al., 2015). We have used the concept of "safe" AWD (AWD15) in the present study and the findings are consistent with the available published literature. In this study, the above-ground biomass was unaffected among treatments in the wet season. In the dry season, RS-B and WRS plots had higher above-ground biomass than RS-I plots. This result indicates that RS incorporation did not have a stimulatory effect on biomass production of rice.

4.2. Effects of water regime on CH_4 and N_2O emissions

Under flooded rice ecosystems, decreasing levels of dissolved O_2 and redox potential favor methanogens (CH_4-producing bacteria) performing anaerobic microbial decomposition, resulting in the production of CH_4 (Lagomarsino et al., 2016). As CH_4 is the end-product of anaerobic decomposition of complex soil organic matter, its emissions from rice fields can be reduced by: (i) limiting the duration of soil submergence by AWD or mid-season drainage instead of CF, (ii) improving organic matter management by reducing C inputs through residue management, (iii) direct establishment of rice crops either by dry or wet direct seeding, and (iv) proper nutrient management (Zou et al., 2005; Tyagi et al., 2010; Zhang et al., 2012; Sander et al., 2014; Yang et al., 2015; Islam et al., 2018). In the present study, CF in the paddy field led to high CH_4 emissions in the cropping season compared to AWD irrigation. RS-B plots under AWD maintained the average CH_4 emissions flux at a low level and there were no obvious peaks during the rice growing period in the wet season (Fig. 2). The average CH_4 emissions from RS-B and RS-I plots under AWD were reduced by 63% and 39%, respectively, compared to the corresponding RS management practices under CF in the wet season (Table 4). In the dry season, the average CH_4 emissions were reduced by 36% from RS-B and 37% from RS-I plots under AWD compared to the same plots under CF (Table 4), although the two-way interaction between water and RS management was not significant. Our findings are similar to Li et al. (2005); Minamikawa and Sakai (2006); Tyagi et al. (2010), and Lagomarsino et al. (2016), who also reported higher CH_4 emission rates from rice fields that were continuously flooded rather than drained once or several times during the growing season.

AWD irrigation reduces CH_4 emissions by preventing the development of a soil reductive condition (Tyagi et al., 2010). This condition helps improve soil permeability and increase soil redox potential, which adversely influence populations and activities of methanogenic bacteria. These bacteria control CH_4 formation under strict anaerobic conditions during the rice-growing season (Sander et al., 2014; Islam et al., 2018). Drainage of flooded water under AWD results in aerobic soil conditions responsible for suppressing the methanogenesis process and reducing CH_4 emissions (Ali et al., 2013). In Thailand, replacing the traditional practice of CF with AWD has been reported to reduce the CH_4 emissions flux by 49% (Chidthaisong et al., 2018). Minamikawa et al. (2014) reported a 30.5% reduction in CH_4 emissions by using intermittent irrigation (AWD) during the rice-growing period. LaHue et al. (2016) also observed a reduction in annual CH_4 emissions by 57–78% under water-seeded or drill-seeded AWD, with the largest reduction corresponding to the drill-seeded rice. The AWD water-saving technique has been reported to reduce CH_4 emissions by 70–90% during dry season rice cultivation (Watanabe et al., 2013). Although, in this study, RS-B plots under AWD had the lowest average CH_4 emissions ($2.73\ mg\ m^{-2}\ h^{-1}$), it had the highest average N_2O emissions ($0.19\ mg\ m^{-2}\ h^{-1}$) in the wet season (Table 4). However, the total GWP of RS-B plots (3618.72 kg CO_2-eq. ha^{-1}) was slightly lower than the total GWP of RS-I plots (4009.94 kg CO_2-eq. ha^{-1}) under AWD in the wet season (Table 5).

Paddy fields are also considered one of the important sources of atmospheric N_2O emissions (Zhu et al., 2015). N_2O is produced from paddy soil as an intermediate product through the microbial processes of nitrification and denitrification (Malla et al., 2005), which are affected by field water management, fertilizer application, and organic amendments (Zou et al., 2005; Wang et al., 2012). In the present study, water regime did not significantly influence the average flux and the seasonal total N_2O emissions from the paddy field in the wet season, whereas water regime only significantly influenced the average flux of N_2O emissions in the dry season (Tables 4–6). The seasonal total N_2O emissions were slightly lower in the dry season than the wet season, and N_2O emissions increased under AWD only in the dry season. Setyanto et al. (2018) also observed higher N_2O emissions in the wet season than the dry season. Wet but drained soil conditions during the wet season might provide a favorable environment accelerating N_2O production and ultimately increasing N_2O emissions. The nitrification process (oxidation of available N) dominates in aerated soil conditions with more releases of NO than N_2O, whereas the denitrification process (reduction due to low O_2 availability) dominates in submerged soil with more releases of N_2O than NO (Pandey et al., 2014). In the present study, the AWD period lasted from 26 to 60 DAB in the wet season and 31–61 DAB in the dry season, which corresponded to the tillering stage of rice. Both the nitrification and denitrification processes simultaneously occur under AWD irrigation. This condition favors the release of more N_2O through the oxygenated soil pores (Pathak et al., 2002). This might cause higher N_2O emissions from plots under AWD than under CF. Xu et al. (2015) also reported higher N_2O emissions from paddy fields with alternate flooding and drainage compared to continuously flooded fields. In this study, N_2O emission rates peaked during late July and late September from almost all plots in the wet season, whereas similar peaks were observed during late November and late December in the dry season (Fig. 2). These increased peaks in N_2O emissions could be due to urea fertilizer topdressing during these periods. N_2O emissions are usually boosted within seven days after N fertilizer application due to an increased availability of N for soil microbes (Xu et al., 2015).

4.3. Effects of RS management on CH_4 and N_2O emissions

Soil incorporation or field burning of RS has its own agronomic and environmental significance. Incorporation of crop residue into flooded soils accelerates the reduction process in the soil and decreases its redox potential more rapidly. It also provides the primary C substrate for methanogens, triggering CH_4 production (Zou et al., 2005; Minamikawa and Sakai, 2006; Gaihre et al., 2013). During the early growth stage of rice, the emissions of CH_4 are comparatively higher in the flooded soil with crop residue as earlier decomposition of RS favors accumulation of acetate, which is a methanogenic precursor (Glissmann and Conrad, 2000). However, subsequent rice crops could be benefited due to the incorporation of RS as it supplies nutrients, especially K. In the present study, the average flux of CH_4 emissions was greater from RS-I and RS-B plots under CF in both seasons compared to plots without any RS incorporation/burning (Table 4). Our findings are similar to those of Zhang et al. (2017) and Tan et al. (2018), who also reported a considerable increase in CH_4 emissions after incorporation of RS. The average flux of CH_4 emissions was similar among RS management practices under AWD, regardless of the growing seasons. RS provides a relatively labile organic substrate, which can be readily degraded to CH_4 (Yuan et al., 2014; Liu et al., 2015).

Incomplete combustion takes place when RS is openly burned in the field, resulting in the generation of GHG including CH_4 and N_2O (Gadde et al., 2009; Romasanta et al., 2017). In this study, the average flux of N_2O emissions was very low compared to the average flux of CH_4 emissions in both seasons (Table 4). Furthermore, residue management practice did not significantly influence the seasonal total N_2O emissions in both growing seasons. Romasanta et al. (2017) also reported nonsignificant average seasonal N_2O fluxes across different straw treatments (straw retained, partial straw removal, complete straw removal, and straw burned). During field burning, the C:N ratio of crop residue declines, as more C is lost than N (Koga et al., 2016). Knoblauch et al. (2011) also reported a higher loss of C than N from the burning of piled rice husk residue. The magnitude of C and N losses during open-field burning of crop residue and subsequent CH_4 and N_2O emission rates from soil after soil incorporation have been reported to be influenced by various attributes, including the type of crop, moisture content of the crop residue, the burning method, and the weather conditions during burning (Koga et al., 2016). GHG emissions per tonne of RS have been reported at 1714 kg CO_2-eq from open field burning, which is approximately 35% higher than RS incorporation into the soil in northern Thailand (Yodkhum et al., 2018).

4.4. Effects of water regime and RS management on GWP and yield-scaled GWP

In the present study, the total GWP and yield-scaled GWP were higher from RS-I and RS-B plots under CF compared to the same plots under AWD in both growing seasons (Tables 5 and 6). WRS plots had the lowest total GWP and yield-scaled GWP within each water management regime, regardless of the growing seasons. Averaged across two seasons, the total GWP under AWD irrigation was reduced by 26.6% compared to CF, and WRS plots had 35.9% and 31.3% lower total GWP compared to RS-I and RS-B plots, respectively (Table 7). Averaged across two seasons, the yield-scaled GWP under AWD irrigation was reduced by 34% compared to CF, and WRS plots had 36.4% and 28.6% lower total GWP compared to RS-I and RS-B plots, respectively (Table 7). Sander et al. (2014) also reported a 26% and 67% reduction in seasonal total GWP with and without residue incorporation, respectively, under drying and wetting soil condition compared to flooded fallow. The authors also reported higher yield-scaled GWP from a dried paddy field with residue than without residue incorporation. Setyanto et al. (2018) reported a 35% reduction in the yield-scaled GWP compared to CF, and the current results for yield-scaled GWP are in line with their findings. CF stimulated higher CH_4 emissions, which covered around 78% and 81% of the total GWP from RS-I and RS-B plots, respectively, in the wet season, whereas the respective values were 81% and 75% in the dry season (Tables 5 and 6). In contrast, AWD resulted in increased N_2O emissions, especially in the dry season. The contribution of N_2O to the seasonal total GWP ranged between 19% and 22% from RS-I and RS-B plots under CF in the wet season, and between 19% and 25% from RS-I and RS-B plots under CF in the dry season. Sander et al. (2014) also observed lower N_2O emissions during the rice growing season where the largest contribution (> 90%) to the cumulative GWP was from CH_4 emissions, irrespective of fallow and residue management practices. In a typical Korean paddy soil, CF-induced CH_4 emissions were responsible for more than 95% of the total GWP and the contribution of N_2O to the total GWP was less than 5% (Kim et al., 2014).

5. Conclusions

Water and crop residue management are two important mitigation strategies to reduce GHG emissions from flooded rice fields. AWD irrigation significantly reduced irrigation water inputs compared to CF in both the wet and dry growing seasons; however, water savings was more in the dry season. AWD also resulted in higher grain yields compared to CF in both growing seasons. The seasonal total CH_4 emissions were very high compared to the seasonal total N_2O emissions in both seasons, irrespective of the water regimes and RS management practices. The total GWP (based on CH_4 and N_2O emissions) and yield-scaled GWP under AWD are lower than that of CF. RS addition either as soil incorporation or open-field burning had little or no effect on irrigation water inputs and grain yields; however, the total GWP and yield-scaled GWP were higher from plots with RS incorporation or burning, in comparison with plots where no RS was incorporated. Therefore, it can be concluded that among the water regime and RS management practices tested in the study, AWD irrigation without any RS application (soil incorporation or burning) could be a potential option for maintaining the yield-scaled GWP of irrigated lowland double-rice cropping at low levels. However, the long-term effects of water regime and RS management practices on rice yields, irrigation water savings, and yield-scaled GWP need to be further evaluated under diverse soil and weather conditions.

Table 7
Changes of total global warming potential (GWP) and yield-scaled GWP from the paddy field during the wet and dry rice growing seasons of 2016–2017 as affected by each of two water regimes (continuous flooding [CF], alternate wetting and drying [AWD]) and three rice straw (RS) management practices (RS incorporation [RS-I], RS burning [RS-B], without RS incorporation and burning [WRS]).

Factor	Reduction of total GWP under AWD over CF (%)		Total GWP from two seasons	Reduction of total GWP from two seasons under AWD over CF (%)	Reduction of yield-scaled GWP under AWD over CF (%)		Yield-scaled GWP from two seasons	Reduction of yield-scaled GWP from two seasons under AWD over CF (%)
	Wet season	Dry season			Wet season	Dry season		
Water management (W)								
CF			8934.04a				2.35a	
AWD			6556.78b	26.6			1.55b	34.0
RS management (R)								
RS-I			9022.45a				2.31a	
RS-B			8426.33a				2.06a	
WRS			5787.45b	35.9 compared to RS-I; 31.3 compared to RS-B			1.47b	36.4 compared to RS-I; 28.6 compared to RS-B
W × R								
CF + RS-I			10481.54				2.83	
CF + RS-B			10417.04				2.57	
CF + WRS			5903.54				1.64	
AWD + RS-I	27.2	28.6	7563.36	27.8	37.3	35.7	1.79	36.5
AWD + RS-B	44.6	27.5	6435.62	38.2	46.6	29.2	1.54	40.1
AWD + WRS	11.5	−9.7	5668.36	3.9	30.3	1.8	1.31	20.7
F-value								
W			19.17**				35.08**	
R			13.41**				13.28**	
W × R			4.22ns				3.02ns	

In a column, numbers followed by same letters are not significantly different by Tukey's honest significant difference test at $P < 0.05$. ns: not significant.
** indicates significance at $P < 0.01$.

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